Table 1 shows the solubility of a range of compounds.

In addition, most ammonium (NH^{+}) compounds are soluble.

Anions	Soluble	Slightly soluble	Insoluble
hydroxides	Group I, Ba^{2+}	Ca^{2+}, Sr^{2+}	all others
carbonates	Group I except Li^{+}	Li^{+}	all others
hydrogen carbonates	most		all others
halides	most	Pb2+ (hot)	Pb^{2+}, Ag^{+}
nitrates	all		all others
sulfates	most	Ca^{2+}, Ag^{+}	Ba^{2+}, Pb^{2+}
sulfides	Groups I & II		all others
sulfites	Group I		all others
phosphates	Group I, except Li^{+}		all others

Table 1

1 Name each compound and describe its solubility in water.

- a $CuSO_4$
- b $CaCO_3$
- c AgCl
- d NH_4OH
- e $Pb(NO_3)_4$
- f Li_3PO_4

Precipitates

Some ionic compounds are soluble in water. When they dissolve, they break up into aqueous ions, which are usually metal ions with a positive charge (cations) and non-metals or radicals with a negative charge (anions). In solution, cations and anions may react together to form insoluble substances called **precipitates** (*precipitate*: to separate and 'rain down'). When an ionic substance dissolves in water, the anions and cations move about in solution, constantly interacting. (Remember opposite electrical charges attract.) If two ions of opposite charge come together and react to form an insoluble substance then a precipitate is formed. When precipitates are formed the reaction is almost instantaneous. To see if a precipitate is formed, and if so, what it is called:

- work out what ions are in the solution
- work out possible combinations of ions
- identify the precipitate formed.

EXAMPLE

i **When a silver nitrate ($AgNO_3$) solution is mixed with a sodium chloride (NaCl) solution a white precipitate forms. What is this precipitate? Write a net ionic equation for the reaction.**

Work out the possible results of positive ions pairing with negative ions: the two possible compounds are silver chloride (AgCl) and sodium nitrate ($NaNO_3$). A table of solubilities (see table 1) tells that silver chloride (a silver halide) is insoluble and sodium nitrate is soluble. Hence the precipitate must be silver chloride.

Show all the ions present:

$$Ag^{+}\,(aq) + NO_3^{-}\,(aq) + Na^{+}\,(aq) + Cl^{-}\,(aq) \rightarrow AgCl\,(s) + Na^{+}\,(aq) + NO_3^{-}\,(aq)$$

Rewrite, omitting the 'spectator ions': $Ag^{+}\,(aq) + Cl^{-}\,(aq) \rightarrow AgCl\,(s)$

2 For each pair of solutions, indicate if a precipitate will be formed and write an ionic equation for the relevant reaction. Include the name of any precipitate formed.

- a Barium chloride ($BaCl_2$) and sodium nitrate ($NaNO_3$).
- b Calcium chloride ($CaCl_2$) and lead (II) nitrate ($Pb(NO_3)_2$).
- c Copper (II) sulfate ($CuSO_4$) and barium hydroxide ($Ba(OH)_2$).

3 Outline how you could remove the precipitate from a mixture.

Calculate the theoretical yield of a precipitate, either in moles or grams: First identify the precipitate and write a net ionic equation. Then:

- balance the equation
- look at the ratio of reactants and products in the precipitation reaction
- look at the question to decide what quantities are needed
- carry out the calculations.

ISBN 978 1 92561 066 6
© Peter Gribben, Wendy St Germain and Blake Education 2005

EXAMPLE

ii What is the name, formula and number of moles of the precipitate formed between an excess of barium chloride solution and 100 mL of 0.5 mol L^{-1} sodium sulfate solution?

- $BaCl_2$ gives Ba^{2+} and $2 \times Cl^-$. Na_2SO_4 gives $2 \times Na^+$ and $1 \times SO_4^{2-}$.
- Combinations include sodium chloride and barium sulfate.
- NaCl is soluble and $BaSO_4$ is insoluble, hence $BaSO_4$ is the precipitate.
- Ba^{2+} (aq) + SO_4^{2-} (aq) → $BaSO_4$ (s) (Spectator ions omitted)
- Ratio is 1:1, hence the number of moles of barium sulfate formed will equal the number of moles of sodium sulfate added.

Find the amount of barium sulfate formed:

- Number of moles = concentration (moles per litre) × volume (litres)
  = 0.5 × 0.1 = 0.05 moles

4 Give the name, formula and number of moles of precipitate formed between each pair of solutions.

a 50 mL of 0.1 mol L^{-1} silver nitrate ($AgNO_3$) is mixed with an excess of barium chloride ($BaCl_2$).

b 100 mL of 2 mol L^{-1} sodium hydroxide (NaOH) is mixed with an excess of a solution of iron (II) sulfate ($FeSO_4$).

Calculate the mass of precipitate

EXAMPLE

iii 40 mL of 0.33 mol L^{-1} potassium bromide (KBr) solution was added to an excess of lead (II) nitrate ($Pb(NO_3)_2$) solution. Give the name, formula and mass of the precipitate formed.

- The possible new compounds formed are potassium nitrate (soluble) and lead (II) bromide (insoluble); hence a precipitate of lead (II) bromide ($PbBr_2$) is formed.
- Net ionic equation: Pb^{2+} (aq) + 2 Br^- (aq) → $PbBr_2$ (s)
- Stoichiometry of reaction indicates that the ratio of moles of potassium bromide present to moles of lead bromide formed is 1:2.
  (Stoichiometry deals with the relationships between quantities of reactants and products.)
- Calculating moles of potassium bromide present: 40 ÷ 1000 × 0.33 = 0.0132 moles. This means that there are 0.0066 moles of lead bromide formed.
- Mass of lead bromide = molar mass × number of moles
- Molar mass of lead bromide = (Pb × 1) + (Br × 2) = (207.2 × 1) + (79.9 × 2) = 207.2 + 159.8 = 367.0 g
- Mass of lead bromide = 367.0 × 0.0066 = 2.422 g

5 Give the name, formula and mass of precipitate formed between each pair of solutions.

a 40 mL of 0.5 mol L^{-1} sodium carbonate (Na_2CO_3) solution mixed with excess magnesium sulfate ($MgSO_4$) solution.

b 80 mL of 0.2 mol L^{-1} potassium sulfate (K_2SO_4) solution mixed with excess barium chloride ($BaCl_2$) solution.

6 Define each term.

a anion

b cation

c precipitate

© Peter Gribben, Wendy St Germain and Blake Education 2005
ISBN 978 1 92561 066 6